CHEONGJAE LEE
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EDUCATION

BOSTON UNIVERSITY — Boston, MA
Master's in Electrical Engineering — Expected Graduation: 12/27
Main Focus of study: ***Cyber-Physical Systems, Robotics, Computer Vision***

THE PENNSYLVANIA STATE UNIVERSITY — State College, PA
Bachelor of Science in ***Electrical Engineering****, Minor in Engineering Mechanics* — 12/23

PROJECTS

MANIMO: GRIP-FORCE-LIMITED GRASPING FOR A MOBILE MANIPULATOR — Boston University
EC601 Team Project, team of 2 (in progress) — Fall 2026 – Present

- Added grip-force limits to an Isaac Lab lift task (per-object force classes, jaw contact sensors, crush penalty) and trained PPO policies on 4,096 parallel robots on BU's GPU cluster.
- Reached an 86% lift rate at a 10–12% crush rate for two force classes in simulation and showed exploration noise had overstated crush rate by up to 17 points.
- Built RGB-D perception for object class, position, size, shape and heading (< 1 mm error at 0.65 m in sim).
- Ran Isaac Sim on BU's GPU cluster in containers and brought up the real LanderPi over ROS 2.

MULTI-CROP DETECTION FOR AN AGRICULTURAL ROBOT — Boston University
Robotics Club, Agrobot Team (in progress) — Fall 2026 – Present

- Fine-tuned a COCO-pretrained Faster R-CNN (ResNet-50 FPN) in PyTorch on LaboroTomato (804 images, 10,610 labels), reaching 0.90 precision and 0.91 recall (IoU > 0.5) on validation.
- Converted 10,610 polygon labels to boxes with zero loss, and ran ablations on input size, NMS and augmentation.

PENN STATE ADVANCED VEHICLE TEAM (AUTODRIVE CHALLENGE II) — State College, PA
Head of Vehicle Systems (ME496) and Electrical Engineer (ME440W Capstone) — 01/23 – 12/23

- Designed the electrical power distribution, sensor suite, and roof rack for an autonomous Chevrolet Bolt EUV.
- Led vehicle-systems hardware: waterproofed sensors and harnesses, and integrated sensors with the central compute.

PROFESSIONAL EXPERIENCE

SOELECT INCORPORATED — Greensboro, NC
Manufacturing Process Engineer — 03/26 – 05/26

- Ran roll-to-roll lithium metal anode production; characterized cells with SEM/EDX, EIS, and cycling.
- Assembled ~5 mAh pouch cells and tested anodes with optical microscopy, contact-angle, and peel-strength tests.

FASTLION ENERGY INCORPORATED — State College, PA
Electrical Engineer — 07/24 – 03/26

- Designed and prototyped lithium-ion Battery Management Systems for 6T military and industrial batteries, covering cell monitoring, fault detection, and system-level protection.
- Designed multi-layer PCBs with PWM control, precision analog sensing, SPI/I2C, and high-current power paths.
- Developed and validated CAN protocols for battery telemetry, diagnostics, and integration with vehicle controllers.
- Brought up, debugged, and validated prototype hardware with oscilloscopes, logic analyzers, and bench instruments.
- Worked with mechanical and systems engineers so designs met performance, safety, and environmental requirements.

EC POWER GROUP — State College, PA
Electrical Engineer (promoted from Electrical Engineering Intern, 02/24) — 05/23 – 07/24

- Led end-to-end development of a battery test platform: PCB design, embedded C/C++ firmware, and DAQ.
- Programmed microcontroller-based BMS prototypes and built firmware and PCBs for an automated battery test rig.

SKILLS

Programming: Python, C/C++, MATLAB, LabVIEW
Robotics and ML: ROS 2, Isaac Sim, Isaac Lab, reinforcement learning (PPO), PyTorch, RGB-D perception
Tools and Hardware: Linux, Git, HPC (Singularity), Altium, KiCad, CAN, SPI, I2C, oscilloscopes, logic analyzers
Lab and Mechanical: SEM/EDX, EIS, li-ion battery cycle testing, SolidWorks, OnShape, 3D printing